B.5 Calibration With a Known Censoring Mechanism: `Optimized` Mode

In this section, we detail the optimized adaptive budgeting algorithm. First, observe that if the budget is sufficiently large such that the sum of $\hat{f}_{\tau_{\text{prior}}}(X_i)$ does not exceed the budget, i.e., $\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i) \leq B$, then we can set the censoring times to the target value $C_i = \hat{f}_{\tau_{\text{prior}}}(X_i)$ for all calibration points. In this case, the solution for (10) is straightforward, and the optimized probabilities are equal to one: $\pi_i^* = 1,\ \forall i \in \mathcal{I}_2$. Since this setup is trivial, we will consider the more challenging setting, in which the target values exceed the threshold $\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i) > B$.

Since each summand $1/\pi_i$ is strictly convex in π_i on $(0,1]$ and the constraint is linear, (10) is a convex program. Furthermore, if the sum of $\hat{f}_{\tau_{\text{prior}}}(X_i)$ exceeds the budget, the constraint turns into an equality constraint, as stated next.

Proposition B.1. *Assume $\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i) > B$. Then any minimizer π^* of (10) must satisfy the budget constraint with equality:*

$$\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i)\,\pi_i^* \;=\; B.$$

We now turn to show how to compute the optimized probabilities π^*. Under Proposition B.1, the optimization constraint in (10) turns into an equality constraint, and the Lagrangian is therefore given by

$$\mathcal{L}(\pi,\lambda) = \frac{1}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2}\frac{1}{\pi_i} + \lambda\Big(\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i)\,\pi_i - B\Big),$$

where $\lambda > 0$ is the Lagrange multiplier. On the open domain $(0,1]^n$, the objective is strictly convex and differentiable, so the single stationary point satisfying the first-order conditions is in fact the unique global minimizer. Taking partial derivatives,

$$\frac{\partial\mathcal{L}}{\partial\pi_i} = -\frac{1}{|\mathcal{I}_2|\,\pi_i^2} + \lambda\,\hat{f}_{\tau_{\text{prior}}}(X_i) \;=\; 0 \quad\Longrightarrow\quad \pi_i = \frac{1}{\sqrt{|\mathcal{I}_2|\,\lambda\,\hat{f}_{\tau_{\text{prior}}}(X_i)}}.$$

Applying the box constraint $\pi_i \leq 1$ gives

$$\pi_i^*(\lambda) = \min\Big\{1,\ \frac{1}{\sqrt{|\mathcal{I}_2|\,\lambda\,\hat{f}_{\tau_{\text{prior}}}(X_i)}}\Big\}.$$

Now, define the budget-usage function

$$U(\lambda) \;=\; \sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i)\,\pi_i^*(\lambda).$$

The above equation has a unique solution $\lambda^* > 0$ which satisfies $U(\lambda^*) = B$. This $\lambda^* > 0$ can be recovered by bisection, across the following interval:

$$\left[\lambda_{\text{low}} = \frac{1}{|\mathcal{I}_2|\max_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i)},\ \lambda_{\text{high}} = \frac{|\mathcal{I}_2|\max_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i)^2}{B^2\min_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i)}\right]. \tag{11}$$

This procedure is summarized in Algorithm 8 and it produces a valid solution, as stated in the following proposition.

Proposition B.2. *The output π^* of Algorithm 8 is the unique solution of (10) which satisfies $\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\text{prior}}}(X_i)\,\pi_i^* \leq B$.*

Algorithm 7 Calibration with a known censoring mechanism: optimized budget allocation

Require: Calibration data $\{X_i\}_{i\in\mathcal{I}_2}$, generative model $\mathcal{G}(\cdot)$, audit function $\texttt{Audit}(\cdot)$, pre-trained quantile regression model $\{\hat{q}_\tau(\cdot)\}_{\tau\in\mathcal{T}}$, target miscoverage rate α, prior quantile τ_{prior}, quantile trimming threshold M, total budget B.

$\{\hat{f}_\tau(X_i)\}_{\tau\in\mathcal{T},i\in\mathcal{I}_2} \leftarrow \{\min(\hat{q}_\tau(X_i), M)\}_{\tau\in\mathcal{T},i\in\mathcal{I}_2}$ // trim the quantile est. (Section 6.1)
$\{\pi_i\} \leftarrow$ Algorithm 8 applied with $\{\hat{f}_{\tau_{\text{prior}}}(X_i)\}_{i\in\mathcal{I}_2}$ // optimized per-prompt evaluation probability (Section 5.2)
$\{(\tilde{T}_i, C_i)\}_{i\in\mathcal{I}_2} \leftarrow$ Algorithm 3 applied to $\{X_i\}_{i\in\mathcal{I}_2}$ with $\{\pi_i\}_{i\in\mathcal{I}_2}$ and $\{\hat{f}_{\tau_{\text{prior}}}(X_i)\}_{i\in\mathcal{I}_2}$
$w(\{X_j\}_{j\in\mathcal{I}_2}, i) \leftarrow \frac{1}{\pi_i},\ i \in \mathcal{I}_2$
Obtain $\hat{L}(x)$ from Algorithm 1, applied with $\{X_i, C_i, \tilde{T}_i\}_{i\in\mathcal{I}_2}$, $\{w_i,\}_{i\in\mathcal{I}_2}$, $\{\hat{f}_\tau(\cdot)\}_{\tau\in\mathcal{T}}$, α, and τ_{prior}.
return $\hat{L}(x)$, the lower predictive bound (LPB) for a test point $X_{\text{test}} = x$.

Algorithm 8 Get Optimal Per-Prompt Evaluation Probabilities

Require: effective sample cost $\{\hat{f}_{\tau_{\text{prior}}}(X_i)\}_{i\in\mathcal{I}_2}$, budget B, bijection tolerance ϵ, initial $\lambda_{low}, \lambda_{high}$

Ensure: Optimal π^* and multiplier λ^* satisfying $\pi_i^* = \min\{1, 1/\sqrt{\lambda^* \hat{f}_{\tau_{\text{prior}}}(X_i)}\}$ and $\sum_{i\in\mathcal{I}_2} \hat{f}_{\tau_{\text{prior}}}(X_i)\,\pi_i^* = B$.

if $B > \sum_{i\in\mathcal{I}_2} \hat{f}_{\tau_{\text{prior}}}(X_i)$ **then**
  $\pi^* \leftarrow [1, \ldots, 1]$, $\lambda^* \leftarrow \texttt{Null}$
else
  while $\sum_{i\in\mathcal{I}_2} \hat{f}_{\tau_{\text{prior}}}(X_i)\,\pi_i^*(\lambda_{high}) > B$ **do**
    $\lambda_{\text{high}} \leftarrow 2\,\lambda_{\text{high}}$
  end while
  $\lambda^* \leftarrow$ bisection on $[\lambda_{\text{low}}, \lambda_{\text{high}}]$ to solve $\sum_{i\in\mathcal{I}_2} \hat{f}_{\tau_{\text{prior}}}(X_i)\,\pi_i^*(\lambda) = B$ within ϵ
  $\pi_i^* \leftarrow \min\{1, 1/\sqrt{\lambda^* \hat{f}_{\tau_{\text{prior}}}(X_i)}\}$ for all $i \in \mathcal{I}_2$
end ifreturn (π^*, λ^*)

C A Deeper Look Into $\hat{\alpha}(\tau)$

In Section 4, we introduced the miscoverage estimator $\hat{\alpha}(\tau)$. In this section, we explore its statistical properties in detail. We begin by deriving its expectation and variance:

Proposition C.1 (Unbiasedness and conditional variance of the weighted miscoverage estimator)**.** *Under the conditional independence assumption 2.1 in Section 2, the estimator $\hat{\alpha}(\tau)$ defined in equation (6) satisfies*

$$\mathbb{E}\big[\hat{\alpha}(\tau)\big] = \frac{1}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2} \mathbb{P}\big[T_i < \hat{q}_\tau(X_i)\big].$$

If we further assume that for all $i \neq i' \in \mathcal{I}_2$, $(C_i, T_i) \perp\!\!\!\perp (C_{i'}, T_{i'}) \mid \{X_j\}_{j\in\mathcal{I}_2}$, we have

$$\mathrm{Var}\big[\hat{\alpha}(\tau) \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big] = \frac{1}{|\mathcal{I}_2|^2}\sum_{i\in\mathcal{I}_2}\Big\{w(\{X_j\}_{j\in\mathcal{I}_2}, i)\,\mathbb{P}\big[T_i < \hat{q}_\tau(X_i) \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big] - \big[\mathbb{P}\big[T_i < \hat{q}_\tau(X_i) \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big]\big]^2\Big\}.$$

See Appendix D.1 for the proof. We remark that the conditional independence and pair-wise independence assumptions in Proposition C.1 hold for all variants of our method from Sections 5. This proposition shows that $\hat{\alpha}(\tau)$ is an unbiased miscoverage estimator, resulting in a useful approximation for the miscoverage estimation variance. Under a simplified assumption that, for a fixed τ, each calibration point $i \in \mathcal{I}_2$ is miscovered at the same constant rate conditional on $\{X_j\}_{j\in\mathcal{I}_2}$, we have the following result.

Proposition C.2 (Variance linearly increasing in mean weight under constant miscoverage)**.** *Under Assumption 2.1, and additionally assuming that for all $i \neq i'$, $(C_i, T_i) \perp\!\!\!\perp (C_{i'}, T_{i'}) \mid \{X_j\}_{j\in\mathcal{I}_2}$, and that for any fixed τ,*

each calibration point $i \in \mathcal{I}_2$ is miscovered at the same constant rate conditional on $\{X_j\}_{j\in\mathcal{I}_2}$,

$$\mathbb{P}\big[T_i < \hat{q}_\tau(X_i) \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big] = \mathrm{Const}_\tau.$$

Then, the variance of $\hat{\alpha}(\tau)$ from (6) is

$$\mathrm{Var}\big[\hat{\alpha}(\tau) \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big] = \frac{\mathrm{Const}_\tau}{|\mathcal{I}_2|}\overline{w}_\tau - \frac{\mathrm{Const}_\tau^2}{|\mathcal{I}_2|},$$

where $\overline{w}_\tau = |\mathcal{I}_2|^{-1}\sum_{i\in\mathcal{I}_2} w_\tau(\{X_j\}_{j\in\mathcal{I}_2}, i)$ is the mean weight.

See proof in Appendix D.2.

D Proofs

D.1 Proof of Proposition C.1

Proof. Recall that

$$\hat{\alpha}(\tau) = \frac{1}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2} w(X_i)\,\mathbb{I}\{\hat{q}_\tau(X_i) \le C_i\}\,\mathbb{I}\{T_i < \hat{q}_\tau(X_i)\}.$$

Set

$$V_i = w(X_i)\,\mathbb{I}\{\hat{q}_\tau(X_i) \le C_i\}\,\mathbb{I}\{T_i < \hat{q}_\tau(X_i)\}.$$

By Assumption 2.1 each of C_i and T_i is independent of the other given X_i. Hence for each i,

$$\begin{aligned}
\mathbb{E}\big[V_i \mid \{X_j\}_{j\in\mathcal{I}_2}\big] &= \mathbb{E}\big[w(\{X_j\}_{j\in\mathcal{I}_2}, i)\,\mathbb{I}\{\hat{q}_\tau(X_i) \le C_i\}\,\mathbb{I}\{T_i < \hat{q}_\tau(X_i)\} \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big] \\
&= \mathbb{E}\big[w(\{X_j\}_{j\in\mathcal{I}_2}, i)\,\mathbb{I}\{\hat{q}_\tau(X_i) \le C_i\} \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big]\,\mathbb{E}\big[\mathbb{I}\{T_i < \hat{q}_\tau(X_i)\} \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big] \\
&= w(\{X_j\}_{j\in\mathcal{I}_2}, i)\,\mathbb{P}\big[\hat{q}_\tau(X_i) \le C_i \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big]\,\mathbb{P}\big[T_i < \hat{q}_\tau(X_i) \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big] \\
&= \mathbb{P}\big[T_i < \hat{q}_\tau(X_i) \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big],
\end{aligned}$$

where the first is by the tower property, and in the last transition we used $w(\{X_j\}_{j\in\mathcal{I}_2}, i) = \mathbb{P}\big[\hat{q}_\tau(X_i) \le C_i \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big]^{-1}$, from (5).

Consequently,

$$\begin{aligned}
\mathbb{E}[\hat{\alpha}(\tau)] &= \mathbb{E}\Big[\tfrac{1}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2} V_i\Big] \\
&= \mathbb{E}\Big[\tfrac{1}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2} \mathbb{E}\big[V_i \mid \{X_j\}_{j\in\mathcal{I}_2}\big]\Big] \\
&= \mathbb{E}\Big[\tfrac{1}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2} \mathbb{P}\big[T_i < \hat{q}_\tau(X_i) \mid X_i\big]\Big] = \frac{1}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2} \mathbb{P}\big[T_i < \hat{q}_\tau(X_i)\big].
\end{aligned}$$

For the conditional variance, we similarly derive

$$\begin{aligned}
\mathbb{E}\big[V_i^2 \mid \{X_j\}_{j\in\mathcal{I}_2}\big] &= \mathbb{E}\big[w_\tau^2(\{X_j\}_{j\in\mathcal{I}_2}, i)\,\mathbb{I}\{\hat{q}_\tau(X_i) \le C_i\}\,\mathbb{I}\{T_i < \hat{q}_\tau(X_i)\} \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big] \\
&= w_\tau^2(\{X_j\}_{j\in\mathcal{I}_2}, i)\,\mathbb{P}\big[\hat{q}_\tau(X_i) \le C_i \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big]\,\mathbb{P}\big[T_i < \hat{q}_\tau(X_i) \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big] \\
&= w_\tau(\{X_j\}_{j\in\mathcal{I}_2}, i)\mathbb{P}\big[T_i < \hat{q}_\tau(X_i) \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big].
\end{aligned}$$

Therefore

$$\begin{aligned}
\mathrm{Var}\big[V_i \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big] =& \mathbb{E}\big[V_i^2 \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big] - \big(\mathbb{E}[V_i \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}]\big)^2 \\
=& w(\{X_j\}_{j\in\mathcal{I}_2}, i)\,\mathbb{P}\big[T_i < \hat{q}_\tau(X_i) \;\big|\; \{X_j\}\big] - \Big[\mathbb{P}\big[T_i < \hat{q}_\tau(X_i) \;\big|\; \{X_j\}\big]\Big]^2.
\end{aligned}$$

Conditioned on $\{X_j\}_{j\in\mathcal{I}_2}$, the pairs (C_i, T_i) are independent across i, and so for $i \neq j$,

$$\mathrm{Cov}(V_i, V_j \mid \{X_j\}) = 0.$$

Consequently,

$$\begin{aligned}\mathrm{Var}\big[\hat{\alpha}(\tau) \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big] =& \mathrm{Var}\Big[\tfrac{1}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2} V_i \;\Big|\; \{X_j\}_{j\in\mathcal{I}_2}\Big] \\ =& \tfrac{1}{|\mathcal{I}_2|^2}\sum_{i\in\mathcal{I}_2}\Big\{w(\{X_j\}_{j\in\mathcal{I}_2}, i)\,\mathbb{P}\big[T_i < \hat{q}_\tau(X_i) \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big] \\ & - \big[\mathbb{P}\big[T_i < \hat{q}_\tau(X_i) \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big]\big]^2\Big\}\end{aligned}$$

□

D.2 Proof of Proposition C.2

Proof of Proposition C.2. Under the proposition's assumptions, Proposition C.1 shows us that the variance of $\hat{\alpha}(\tau)$ simplifies to

$$\begin{aligned}\mathrm{Var}\big[\hat{\alpha}(\tau) \;\big|\; \{X_j\}_{j\in\mathcal{I}_2}\big] &= \frac{1}{|\mathcal{I}_2|^2}\sum_{i\in\mathcal{I}_2}\Big\{w_\tau(\{X_j\}_{j\in\mathcal{I}_2}, i)\,\mathrm{Const}_\tau - \mathrm{Const}_\tau^2\Big\} \\ &= \frac{\mathrm{Const}_\tau}{|\mathcal{I}_2|}\overline{w}_\tau - \frac{\mathrm{Const}_\tau^2}{|\mathcal{I}_2|}\end{aligned}$$

where $\overline{w}_\tau = |\mathcal{I}_2|^{-1}\sum_{i\in\mathcal{I}_2} w_\tau(\{X_j\}_{j\in\mathcal{I}_2}, i)$ is the mean weight. □

D.3 Proof of Proposition B.1

Proof of Proposition B.1. Since

$$\frac{\partial}{\partial\pi_i}\Big(\frac{1}{|\mathcal{I}_2|}\sum_{j\in\mathcal{I}_2}\tfrac{1}{\pi_j}\Big) = -\frac{1}{|\mathcal{I}_2|\,\pi_i^2} < 0,$$

increasing any single π_i (while keeping the others fixed) strictly decreases the objective. Assume for the sake of contradiction that there exist an optimal solution for (10), π^*, satisfying $\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\mathrm{prior}}}(X_i)\,\pi_i^* < B$. Then, there exist an index $j \in \mathcal{I}_2$ and $\delta > 0$ such that $\pi_j^* + \delta \leq 1$ and $\sum_i \hat{f}_{\tau_{\mathrm{prior}}}(X_i)\,\tilde{\pi}_i \leq B$, where $\tilde{\pi}_i = \pi_i^*$ for $i \neq j$ and $\tilde{\pi}_j = \pi_j^* + \delta$. But then

$$\frac{1}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2}\frac{1}{\tilde{\pi}_i} \;<\; \frac{1}{|\mathcal{I}_2|}\sum_{i\in\mathcal{I}_2}\frac{1}{\pi_i^*},$$

contradicting optimality of π^*. Therefore, the budget must bind:

$$\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\mathrm{prior}}}(X_i)\,\pi_i^* = B.$$

□

D.4 Proof of Proposition B.2

Proof of Proposition B.2. Notice that if $\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\mathrm{prior}}}(X_i) \leq B$, then $\pi_i^* = 1$ for all $i \in \mathcal{I}_2$ is the optimal solution which satisfies the constraint. We now turn to consider the case where $\sum_{i\in\mathcal{I}_2}\hat{f}_{\tau_{\mathrm{prior}}}(X_i) > B$. We first show that there exists a unique solution λ^*. Observe that $U : (0,\infty) \to (0, \sum_i \hat{f}_{\tau_{\mathrm{prior}}}(X_i)]$ is continuous and strictly decreasing, with $\lim_{\lambda\to 0^+} U(\lambda) = \sum_i \hat{f}_{\tau_{\mathrm{prior}}}(X_i) > B$ and $\lim_{\lambda\to\infty} U(\lambda) = 0$. By the intermediate-value theorem there is a unique $\lambda^* > 0$ such that $U(\lambda^*) = B$. Next, the output of Algorithm 8 is indeed the solution λ^* since U is a monotone function, and thus the bisection approach is valid in this case. Therefore, $\pi^*(\lambda)$ is the correct optimal solution. □